

# Elijah Feldman

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## EDUCATION

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- **Thomas Jefferson High School for Science and Technology** Alexandria, VA  
*High School Diploma; GPA: 4.0* Class of 2028
  - **Coursework:** AP Biology, AP Computer Science, Research Statistics 1

## EXPERIENCE

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- **George Mason University** Fairfax, VA  
*Research Intern* June 2024 – Present
  - **LLM Finetuning:** Created a manually annotated 500+ sample dataset for parameter-efficient finetuning of an 8-billion parameter language model. Achieved a 25% performance improvement over a LLaMA 70B baseline in relationship and entity extraction tasks for legal court filings.
  - **High-Performance Computing (HPC):** Ran training pipelines and evaluation metrics on A100 80GB GPU clusters using Slurm to send experiment jobs. Used Ollama for local model deployment and testing.
  - **Empirical Evaluation & Data Analysis:** Evaluated the fine-tuned model against both the model prior to fine-tuning and another baseline over thirty trials. Conducted rigorous statistical analyses to establish normality, correct for covariance, and validate statistical significance across trials.
- **TJ Computer Systems Lab** Alexandria, VA  
*Systems Administrator* Aug 2024 – Present
  - **Infrastructure Management:** Maintained 30+ Debian-based workstations supporting a user base of over 60 students and faculty. Deployed and managed a web-based UI infrastructure enabling remote desktop access for 50+ concurrent users.
  - **Cluster Administration:** Maintained a 50+ node compute cluster. Deployed containerized virtual environments and optimized high-performance compute resources for the neuroscience laboratory.

## PROJECTS

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- **Watermarking Diffusion Language Models:** Independent project mentored by a George Mason University faculty member involving the creation of a robust method for applying a watermark to diffusion language model generated text.
- **BioBlitz:** Built and deployed a full-stack educational web platform serving 300+ active users with over 1000 completed assessments for MCAT or Biology Olympiad preparation. Implemented a recommendation system and Elo-based scoring algorithm. Utilized A/B testing to optimize platform analytics and user engagement.

## AWARDS & HONORS

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- **American Invitational Mathematics Examination (AIME):** Qualifier & AMC 10 Honor Roll (Top 2.5%)
- **United States of America Computing Olympiad (USACO):** Gold Division (Top ~5% globally)
- **United States of America Biology Olympiad (USABO):** 2x Honorable Mention (Top 20% nationally)

## TECHNICAL SKILLS

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- **Programming & Markup:** Python, Java, JavaScript, LaTeX, HTML/CSS
- **ML & Data Libraries:** Hugging Face, scikit-learn, NumPy, pandas
- **Tools & Frameworks:** Linux, Slurm, Ollama, Git, React
- **Core Competencies:** High-Performance Computing (HPC), System Administration, Knowledge Graphs, Statistical Data Analysis

## PUBLICATIONS

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- **FineREX: Fine-Tuned NER-RE for Human Smuggling Knowledge Graphs:** Elijah Feldman, Dipak Meher, Carlotta Domeniconi. Under Review, ICKG 2026. arXiv:2606.19710